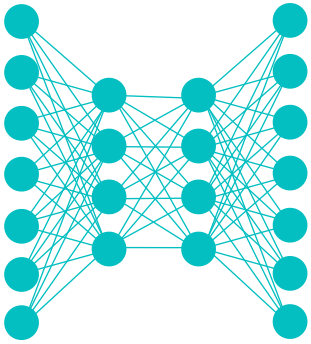
Lecture Notes for

Neural Networks II

CNN Circuits

Next Time:
Fully Convolutional Learning

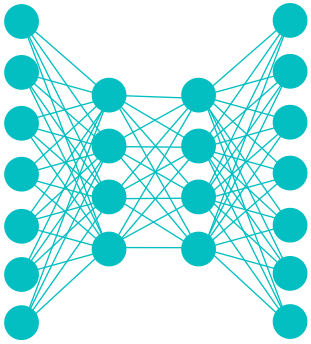
Reading: Chollet 5.4



Lecture Notes for

Neural Networks II

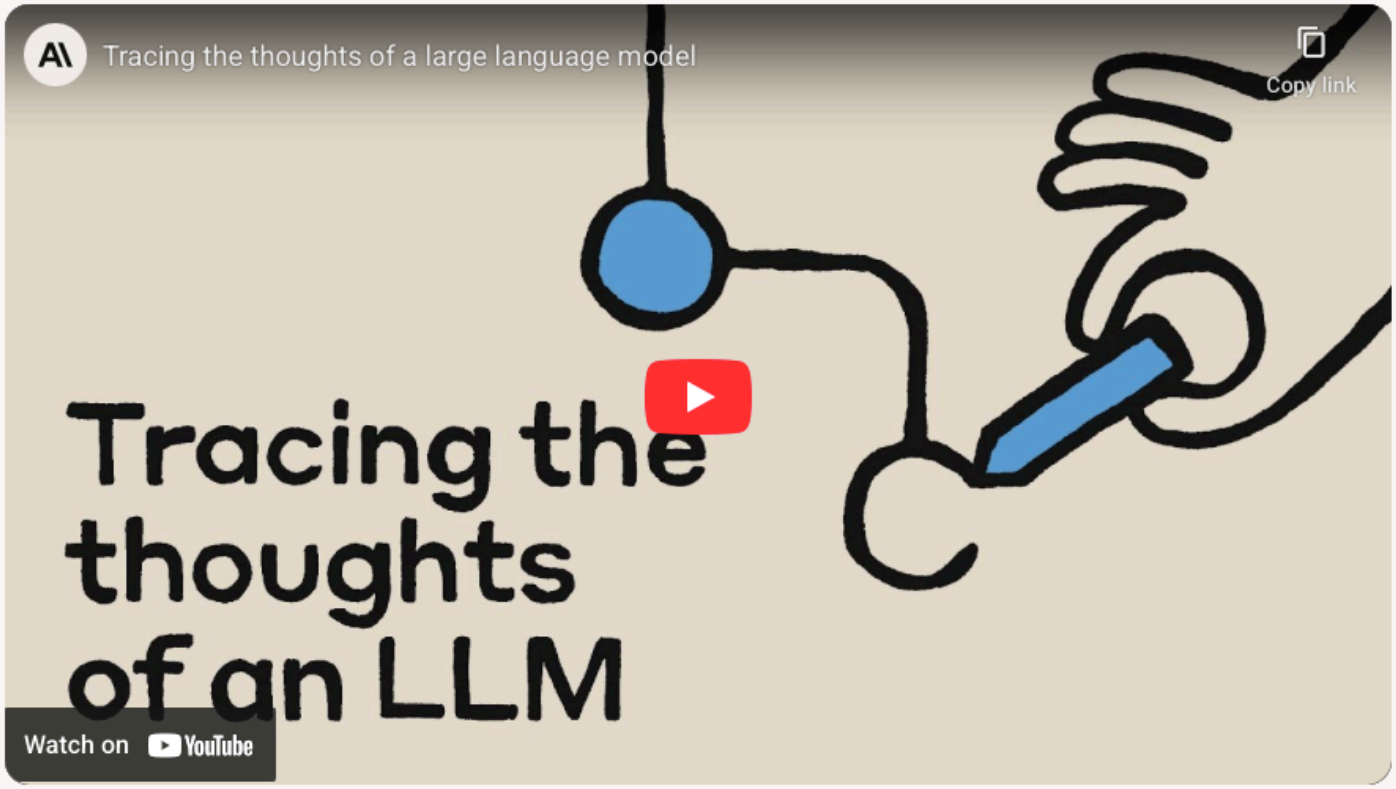
Mid-term Review



Logistics and Agenda

- Logistics
  - Grading Update
- Agenda
  - Last Time: Branch Specialization and Universality
  - Student Presentation
  - Review
  - Next Time:
    - Mid-Term

Student Paper Presentation



Topics on midterm

- Ethical Considerations
  - Guidelines
  - Explicit Loss and Implicit Methods
  - ConceptNet
- Transfer Learning
- Transformers
  - Overview, Efficient Attention
  - Positional Encoding (RoPE)
  - Encoder Models (BERT), GPT, LLMs
- State Space Models
  - General SSMs
  - Mamba Variants

Papers to Review at High Level

- FAME: Equitable EHR
  - Multiplicative modality weighting for fairness
- MITRA
  - Sampling methods for TabPFN
- Softmax Taylor Expansion
  - Kroenecker product of increasing recurrent state
- Sensor Transformers
  - hierarchical embedding and separate attention
- DroPE
  - Dropping PE, what is the alternative?
- Tiny Recursive Models
  - Feedback loops on tiny models

Example Long Format Question

- Recall the ethical guidelines discussed in class (see below). Which guidelines are violated from the following (and why). How might each violation be mitigated?
  Using generative AI to create video content for YouTube, based upon current trends from online searches and popular news. The app is meant to capitalize on advertising revenue from clicks.

Reliability: does system operate in accordance with intended purpose? **Fairness**: will system be inclusive and accessible? Will it involve or result in unfair discrimination against individuals, communities, or groups?	**Model Measurement and Objective Alignment**
Beneficence: does system benefit individuals, society, or environment? **Respect**: does system respect human rights and autonomy of individuals?	**Forethought and Insight**
Privacy: will system respect and uphold privacy rights and data protection, and ensure the security of data? **Transparency**: will system ensure people know when they are engaging with an AI system? Or know if significantly impacted? **Contestable**: will there be a timely process to allow people to challenge the use or output of the AI system?	**Deployment Design**
Accountability: Those responsible for the different phases of the AI system lifecycle should be identifiable and accountable for the outcomes of the AI systems, and *human oversight* of AI systems should be enabled.	**Organizational Structure**

Example Question

- The methods used for training ConceptNet Numberbatch are based on the idea of implicit knowledge injection to reduce bias. How can this same principle be used in the context of Retrieval Augmented Generation (RAG)? Would a RAG question answering system be more or less biased than an LLM question answering system? Why?

Example Question

- Mostly True or Mostly False, why?
  - BERT has documented emergent abilities.
  - Most LLM models start from BERT before using generative pre-training.
  - Positional encoding is a vital part of modern auto-regressive LLMs.
  - BERT Transfer Learning methods only use the final hidden layer output.
  - Linear attention typically outperforms softmax attention in terms of evaluation metrics (accuracy, recall, etc.)
  - Multi query attention has more keys and more queries compared to multi-head attention.

Example

- Recall flash attention tiling. There are some extra parameters to track in the tiles compared to taking softmax of the entire row. Which parameters are these? Circle them.
- What is the computational complexity for Flash attention?

$$\mathbf{x}^{(1)} = [a, b] \quad f(\mathbf{x}^{(1)}) = \left[e^{a-m(\mathbf{x}^{(1)})}, e^{b-m(\mathbf{x}^{(1)})}\right]$$

$$\mathbf{x}^{(2)} = [c, d] \quad f(\mathbf{x}^{(2)}) = \left[e^{c-m(\mathbf{x}^{(2)})}, e^{d-m(\mathbf{x}^{(2)})}\right]$$

Inner Loop

FlashAttention

$$m(\mathbf{x}) = \max\left(m(\mathbf{x}^{(1)}), m(\mathbf{x}^{(2)})\right)$$

Need to track this

$$s_i = e^{m(\mathbf{x}^{(i)})-m(\mathbf{x})}$$

Slightly more FLOPS, but better utilization of parallelism

$$f(\mathbf{x}) = \left[s_1 \cdot f\left(\mathbf{x}^{(1)}\right), s_2 \cdot f\left(\mathbf{x}^{(2)}\right),\right]$$

$$l(\mathbf{x}) = s_1 \cdot l\left(\mathbf{x}^{(1)}\right) + s_2 \cdot l\left(\mathbf{x}^{(2)}\right)$$

Combine distributed ops:

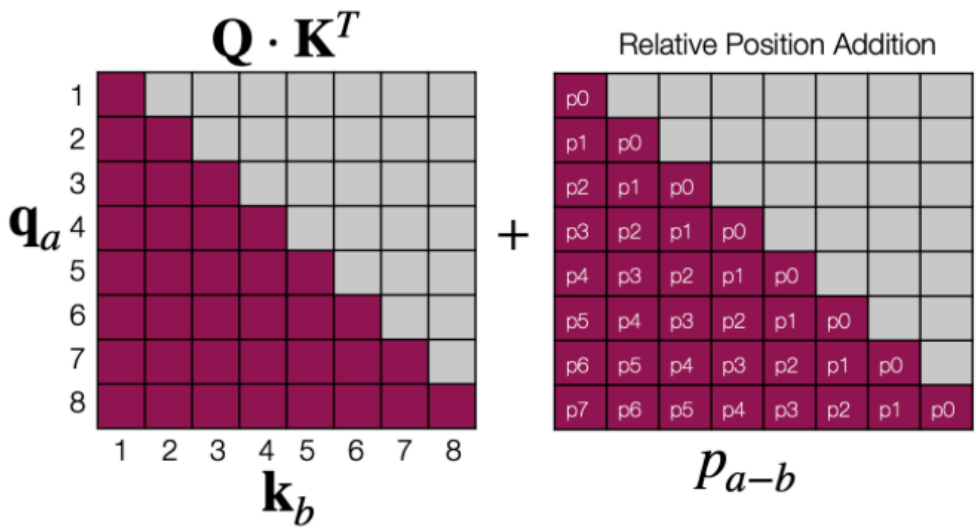
$$\text{softmax}(\mathbf{x}) = \frac{f(\mathbf{x})}{l(\mathbf{x})}$$

Distributed Softmax

55

Example Questions

- Recall relative positional encoding. What are the trainable parameters in the diagram below (or that influence parameters below)?
- How much memory overhead is added to self attention with this method?



Example

- Selective state space models are fundamentally different from transformers because they employ dynamic state matrices. True? False? Why?
- What about Mamba-2?

$$h_t = A_t \cdot h_{t-1} + B_t \cdot x_t \quad \text{and} \quad y_t = C_t^T \cdot h_t$$

Other Questions?